

Smart Electronic Clothing

Clothing will no longer be just a fashion statement. By integrating multiple electronic devices, it will make lives of the wearers much more comfortable

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Adances in textile technology, computer engineering and materials science are promoting a new breed of functional fabrics resulting in some truly smart and clever clothing. Clothing is being developed for protection from chemical, biological and nuclear threats. Besides, with the development of polymers having exotic qualities in terms of their mechanical, electrical and magnetic properties, scientists are ready to design electronic clothing that is capable of heating, cooling and illuminating the body.

Fashion designers are adding wires, circuits and optical fibres to traditional textiles, creating garments that glow in the dark or keep the wearer warm. Meanwhile, electronics engineers are sewing conductive threads and sensors into body suits that can map users' whereabouts and respond to environmental stimuli.

Researchers agree that the development of genuinely interactive electronic textiles is technically possible, and that challenges in scaling up the handmade garments will eventually be overcome. Ideas include the development of jacket-sleeve keypads for controlling cellphones, pagers or MP3 players, and sportswear with integral fabric sensors and display panels to monitor heart rate and blood pressure during a gym workout or morning run. Clothing fitted with textile global positioning system technology could be suitable for locating skiers or mountaineers in bad weather or even for keeping a watch on children.

These smart clothes will be worn



GPS KnowWhereJacket

like ordinary clothing, providing help in various situations according to the designed application.

Steps towards smart clothing

With miniaturisation, consumer electronics manufacturers and chipmakers are managing to squeeze digital music

players, Internet-surfing devices, mobile phones and related devices into smaller and smaller packages.

Some of the wearable designs remain concept prototypes on display only at cyber-fashion shows; others are already, or will soon be, on sale. There is a jacket available that can be plugged into an iPod and used to navigate song